

Homework-7

1. Let $K = k(\alpha_1, \dots, \alpha_n)$ be fields. Prove that any automorphism σ of K that is k -linear, is completely determined by its values $\sigma(\alpha_i), 1 \leq i \leq n$. Show that a k -linear (field) automorphism σ (as above) of K is identity of K iff $\sigma(\alpha_i) = \alpha_i \forall i$.
2. Are $\mathbb{Q}(\sqrt{2})$ & $\mathbb{Q}(\sqrt{3})$ isomorphic as fields?
3. Let $\Phi_n(x) \in \mathbb{Q}[x]$ be the n^{th} cyclotomic polynomial. Find Φ_n for low values of n .
4. Prove the following identities;
 - (i) $\Phi_m(x) = (x^m - 1) / \left(\prod_{\substack{d|m \\ d < m}} \Phi_d(x) \right)$.
 - (ii) $\Phi_p(x) = 1 + x + x^2 + \dots + x^{p-1}$, p a prime.
 $\Phi_{p^n}(x) = \Phi_p(x^{p^{n-1}})$.
 - (iii) For $n = p_1^{r_1} \dots p_s^{r_s}$ (prime factorization),
 $\Phi_n(x) = \Phi_{p_1 \dots p_s}(x^{p_1^{r_1-1} \dots p_s^{r_s-1}})$
 - (iv) For n odd, $\Phi_{2n}(x) = \Phi_n(-x)$.
 - (v) For p a prime not dividing n , $\Phi_{pn}(x) = \frac{\Phi_n(x^p)}{\Phi_n(x)}$.
 - (vi) $\Phi_n(x) = \prod_{d|n} (x^{n/d} - 1)^{\mu(d)}$; μ is Möbius funcⁿ.
 - (vii) $\sum_{d|n} \mu(d) = \begin{cases} 1 & n=1 \\ 0 & \text{if } n>1 \end{cases}$.